

1. To transfer value (6F17) 16-BIT data from microprocessor to output port with address 3F0H the following instructions can be used:

- A. MOV DX, 03F0H; MOV BX, 6F17H; OUT DX, BX
B. MOV DX, 03F0H; MOV AX, 6F17H; OUT DX, AX
C. MOV DX, 03F0H; MOV AX, 6F17H; OUT DX, AL
D. MOV DX, 03F0H; OUT DX, AL

2. The interrupt generated by the execution of an interrupt instruction INT is

- A. NMI interrupt B. software interrupts C. external interrupt D. Interrupt-in-interrupt

3. The physical address of interrupt vector associated with INT 16 is:

- A. 00042H B. 00040H C. 00064H D. 0003CH

4. The size of interrupt vectors table is —

- A. 4 bytes B. 1024 bytes C. 64 bytes D. 128 bytes

5. The binary value of AL and carry flag (CF) after the following instructions have executed :

MOV AL, 0ACH
SAR AL, 1

- A. AL=01010101B; CF=1
B. AL=11010110B; CF=0
C. AL=01010100B; CF=0
D. AL=11010110B; CF=1

6. The binary value of AL and CF after the following instructions have executed is

MOV CL, 4
MOV AL, 7FH
ROL AL, CL

- A. AL=11110111B; CF=0
B. AL=11110111B; CF=1
C. AL=01111111B; CF=1
D. AL=01111111B; CF=0

7. In the following instruction sequence, show the value of AL, in binary:

MOV AL, 3FH
NEG AL
XOR AL, FFH

- A. AL= 00111110B B. AL= 01011111B C. AL= 01011110B D. AL= 01011111B

8. If there is a carry from lowest 4 bit during addition, _____ flag sets.

- A. Auxiliary carry B. Carry C. Over flow D. Sign

9. The result of multiplication instruction MUL BX is stored in register pair

- A. DX:AX B. AX:DX C. CX:AX D. AX:CX

10. Find the content of register AL after execution of the following code:

MOV AL, 11
SHR AL, 1
ADC AL, 1

- A. 7 B. 5 C. 4 D. 3

11. Find the content of memory location X after execution of the following code

X DB 7
MOV AX, 11
MOV BL, 3
DIV BL
OR AH, 30H
MOV X, AH

- A. X='2' B. X=0 C. X='0' D. X=2

12. Find the content of AH registers after execution of the following code:

MOV AX, 14
MOV BL, 4
DIV BL
XOR AH, 0FH

- A. 00H B. C0H C. 0DH D. 0CH

13. LOPPE instruction repeat a group of instruction while

- A. (CX=0) & (ZF=0) B. (CX=0) & (ZF≠0) C. (CX≠0) & (ZF=1) D. (CX≠0) & (ZF=0)

Part 1: MULTIPLE CHOICES QUESTIONS (20 marks)

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
Q11	Q12	Q13	Q14	Q15	Q16	Q17	Q18	Q19	Q20

Part 2: Essay Questions (20 marks)

QUESTION #2

Write instructions to test bit-5 in register AL, if it is one reset bit-1 otherwise set bit-2

Write a procedure to store '+' into 1 KB memory locations defined as below:

X DB 1024 DUP(?)

Using BIOS interrupt, write assembler codes to clear screen and set cursor to position (10, 20).

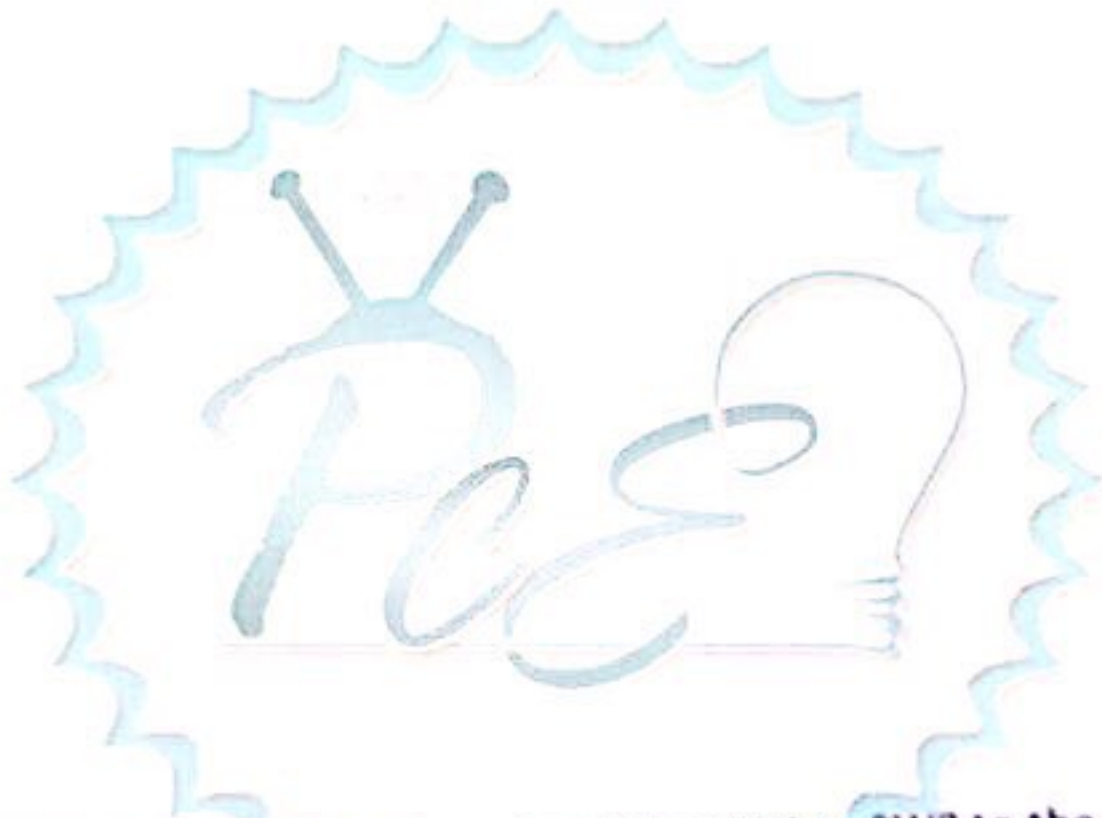
Write an assembler code to reverse string X in order; use LODSB and STOSB strings; where
X DB 'ABCDEK+++'

QUESTION 3

Interface 8-bit input port (74LS245) to read the status of switches SW1 to SW8 to the microprocessor system. Store the status in memory location X defined as byte. Assume the address of the port is 08FFH.

- Draw block circuit diagram circuit, showing address decoding logic, and control signals
- Write a program to read the status of switches and store the result into X memory location

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